Report on SOC Analyst

1. Introduction

A **SOC Analyst (Security Operations Center Analyst)** is a cybersecurity professional responsible for monitoring, detecting, and responding to security incidents and threats within an organization. SOC Analysts are the first line of defense in cybersecurity. The SOC Analyst plays a vital role in identifying and mitigating potential security breaches, ensuring the organization's systems and networks remain secure against cyberattacks.

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2. Work of a SOC Analyst in Jio

In an insightful meeting held with a SOC Analyst, Mr Rohan Lokhande, from Security Operations Center's team, I got the knowledge of a SOC Analyst's work. SOC Analysts typically work within a Security Operations Center (SOC), which is a centralized unit that handles and manages security operations, monitoring, and incident response.

As I spoke with Mr. Rohan Lokhande, I got to understand his basic workflow when an attack is detected. For example, in the case of a brute force attack, alerts are triggered based on rules in the SIEM system. The SOC analyst then checks key details, including the source user, source IP, and destination IP. These are the first checks. They also verify if the source is involved in any ongoing activity, like server hardening. If so, the alert is marked as a "benign positive" or known activity that is not harmful.

If no such activity is found, the SOC analyst raises a ticket for the source IP owner, informing them about the brute force attempt, highlighting the source IP and destination IP. The ticketing platform used is Archer, where tickets are created as "incidents". Once the ticket is created, an email is sent to the server owner with the attack details and evidence, such as screenshots from the SIEM tool (LogRhythm), failed login attempts, and other related logs.

For more serious incidents, such as a malware attack on a file, the file's hash is checked to determine if it's malicious or genuine. If the file is found to be legitimate, the SOC analyst investigates why the alert was triggered and verifies logs and other event details.

The majority of the work involves log analysis and incident response. For critical alerts, SOC analysts consult with higher-level analysts and, when necessary, work with the digital forensics team for deeper investigations.

JSEM (Jio Security Event Management) Platform is a SIEM tool developed by Jio with the help of an open-source tool named, Wazuh. This tool also holds importance in detecting and provides assistance in responding to attacks.

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3. Basic Workflow

Basic workflow of SOC Analysts includes the following steps:

a. Monitoring Security Events

- Constantly monitoring network traffic, logs, and alerts generated by security tools (e.g., SIEM systems, firewalls, intrusion detection systems).

b. Incident Detection and Analysis

- Detecting incidents with the help of various IDS/IPS, firewalls, SIEM, etc.
- Analyzing security alerts and events to detect suspicious or malicious activity.
- Prioritizing incidents based on severity and potential impact, and escalating to appropriate teams if needed.

c. Incident Response

- Taking immediate action to contain and mitigate security threats.
- Coordinating with other teams, such as IT, for eradication of the attack.
- Recovering the system to the state it was in before the attack took place thereby restoring its operational ability.

d. Reporting and Documentation

- Documenting all security incidents, actions taken, and outcomes for compliance and audit purposes and potential attacks in future.
- Generating reports on security events, trends, and vulnerabilities for management and stakeholders.

e. Continuous Improvement

- Continuously improving detection and response processes to stay ahead of evolving threats.
- Having sound knowledge of the current cyberattacks and taking a proactive approach in detection of attacks.
- Awaring the users about the attacks in today’s world through training programs.

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4. Skills

- **Security Information and Event Management (SIEM)**: Knowledge of SIEM platforms like Microsoft Sentinel, LogRhythm, or IBM QRadar for analyzing security logs and events.
- **Networking**: Knowledge of network protocols, TCP/IP, and network security practices such as firewalls, VPNs, and intrusion detection/prevention systems.

- **Operating Systems**: Knowledge of Windows, Linux, and macOS to analyze system logs and understand vulnerabilities.
- **Malware Analysis**: Knowledge of how to analyze and detect malware and other malicious software.
- **Incident Response Tools**: Knowledge of tools like Wireshark, or other forensic analysis tools to investigate security incidents.

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5. Challenges faced by SOC Analysts

SOC Analysts face multiple challenges including:

a. Improper log parsing

Inaccuracy in log parsing could potentially lead to removal of important data, discrepancies in crucial information, inconsistent interpretation of log data, etc. This results in high number of false-positives and enormous amount of alerts.

b. High number of alerts

SOC Analysts may receive high number of alerts at a particular time sometimes amounting to 200-300 in a particular shift thereby overwhelming them. This could lead to fatigue also known as alert fatigue. This could lead to difficulty in prioritizing alerts.

c. High number of false-positives

A false-positive happens when a security system raises an alert for an activity that is actually harmless. These waste time of SOC Analysts thereby contributing towards the team inefficiency. This could again lead to difficulty in differentiating between the more and lesser critical alerts

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6. Tools Used

SOC Analysts use various tools including:

- **SIEM (Security Information and Event Management) :** Splunk, LogRhythm, Microsoft Sentinel and IBM QRadar for analyzing security logs and detecting abnormal activities.
- **Firewalls** and **Intrusion Detection/Prevention Systems (IDS/IPS) :** Snort to monitor traffic and detect unauthorized access attempts.
- **Ticketing tools :** Archer to raise tickets representing incidents
- **Network Traffic Analysis Tools :** Wireshark for packet capture and inspection.

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7. Conclusion

In the discussion with Mr. Rohan Lokhande, I observed that SOC Analysts are key players in cybersecurity. Their role is essential in protecting organizations from cyber threats, ensuring timely detection and response to incidents, and maintaining security standards to safeguard data and infrastructure. By maintaining vigilance, responding mindfully and restoring the system to the normal state, SOC Analysts help organizations mitigate risks and protect against evolving threats.

Prepared By : *Ms. Nikita Suhane*

Consulted (SOC Analyst) : *Mr. Rohan Lokhande*

Manager (SOC Team) : *Mr. Jitendra Tripathi*